

Having this native ability to work with tens of thousands of individual contracts not only brings a higher level of fidelity, but also opens up all kinds of interesting opportunities for creative strategy design.

Since the demise of Quantopian, there have been multiple forks of Zipline by different developers or teams. As of writing this, the most interesting is *Zipline-reloaded*. Free and open source, and ready for you to play with.

■ Sourcing Your Data

A backtester without data is as useless as simile in a trading book and settling on a backtesting software is only half the solution. There's a multitude of providers of financial data, ranging from free to extremely expensive and they vary greatly in coverage, quality, and various technical aspects. If you already work in the financial industry, you probably already have your preferred data sources all lined up, but you may perhaps need something a little different for backtesting futures strategies, in particular, if you decide to take the more realistic route and use Zipline to backtest on individual futures contracts.

It may be tempting to go with some free data source. There's no shortage of websites where you can download plenty of history, free of charge. While there's perfectly fine open source software, I would not particularly trust the quality of free data. If you do use such free sources, make sure that you build in checks in your code for unexpected data spikes, and you might want to use multiple free sources to cross-check against.

Free data is generally more readily available and of higher quality for stocks, in particular, American stocks, but, for futures, it gets a little trickier.

As we mentioned before, there are two related concepts in terms of time series for futures. We have the individual contract series and we have the calculated continuations. What exactly you need from a data provider depends on how you intend to use it and what your software is capable of. Old school software solutions are only really capable of using the continuations. They use a single pre-calculated continuation series for each futures market and thereby greatly cut down on the amount of data processing needed. However, it's not a very realistic way of testing. A better solution would be to use every single individual contract for backtesting. That can easily amount to tens of thousands of individual time-series. If you have a